Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

CODE:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

df = pd.DataFrame({

   'Date': ['01-01-2023', '02-01-2023','03-01-2023','04-01-2023','05-01-2023',],

   'Product': ['Product A', 'Product B','Product A', 'Product C','Product B',],

   'Sales': [200,150,220,300,180,],

   'Quantity': [4,3,5,6,4,],

   'Region': ['North','South','North','East','West',]

})

print(df.head())

print(df.isnull().sum())

df['Sales'] = df['Sales'].fillna(df['Sales'].mean())

df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

print(df.describe())

product_summary = df.groupby('Product').agg({

   'Sales': 'sum',

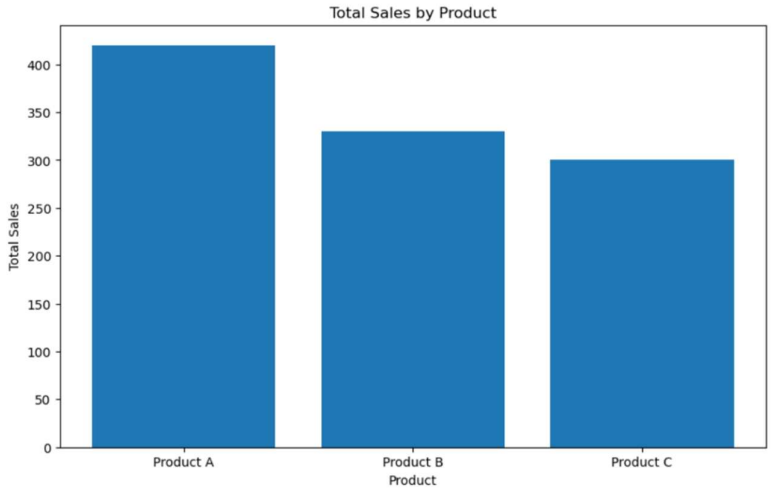
   'Quantity': 'sum'
```

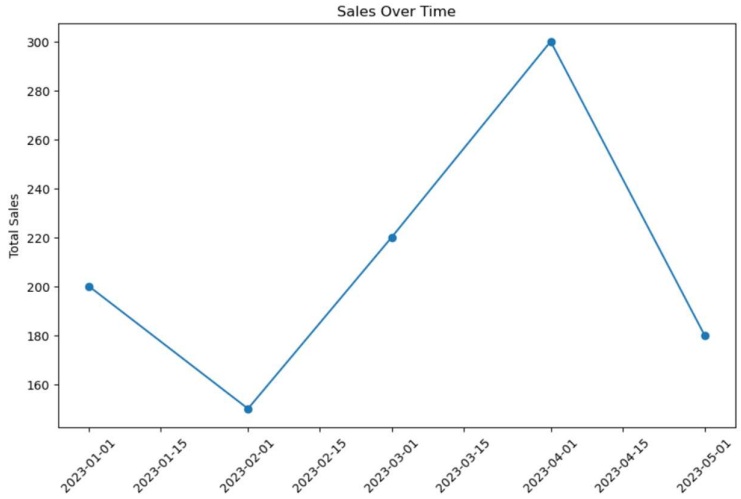
```
}).reset_index()
print(product_summary)
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date')['Sales'].sum().reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'],
marker='o')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.show()
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
```

```
    fill_value=0
)
print(pivot_table)
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)
plt.figure(figsize=(8, 6))
plt.imshow(correlation_matrix, cmap='coolwarm')
plt.colorbar()
plt.xticks(range(len(correlation_matrix.columns)),
correlation_matrix.columns)
plt.yticks(range(len(correlation_matrix.columns)),
correlation_matrix.columns)
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT

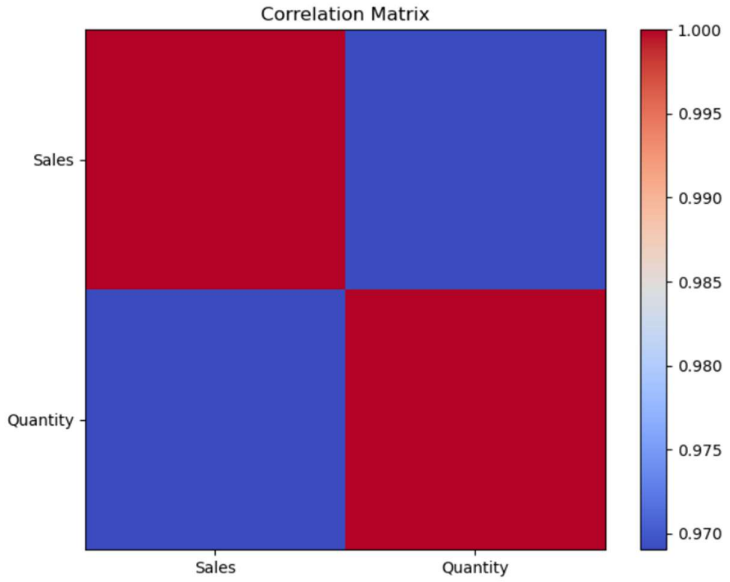
```
         Date    Product  Sales  Quantity Region
0  01-01-2023  Product A    200         4  North
1  02-01-2023  Product B    150         3  South
2  03-01-2023  Product A    220         5  North
3  04-01-2023  Product C    300         6   East
4  05-01-2023  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales  Quantity
count    5.000000  5.000000
mean   210.000000  4.400000
std     56.568542  1.140175
min    150.000000  3.000000
25%    180.000000  4.000000
50%    200.000000  4.000000
75%    220.000000  5.000000
max    300.000000  6.000000
     Product  Sales  Quantity
0  Product A    420         9
1  Product B    330         7
2  Product C    300         6
```





Product	Product A	Product B	Product C
Region			
East	0	0	300
North	420	0	0
South	0	150	0
West	0	180	0

	Sales	Quantity
Sales	1.000000	0.969021
Quantity	0.969021	1.000000



RESULT: Hence data preprocessing and visualization has been performed on the data set given in csv format.